PROJECT FINAL REPORT

Project Title

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

1. INTRODUCTION

1.1 Project Overview

The **Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer** project is a ServiceNow-based automation solution designed to simplify and standardize the process of requesting standard laptops within an organization.

In a traditional IT procurement process, employees may need to submit requests manually, and IT or procurement teams may need to review the request, create fulfillment tasks, assign them to the appropriate team, and update the request status manually. This can result in additional effort, delays, and inconsistent processing.

The proposed solution uses **ServiceNow Service Catalog and Flow Designer** to automate the creation of a laptop procurement task. When an employee selects the **Standard Laptop** catalog item and clicks **Order Now**, the Service Catalog request triggers a Flow Designer workflow. The flow automatically creates a Catalog Task and populates the required information, including the short description, description, assignment group, and approval value.

The main workflow is:

Service Catalog → Standard Laptop → Order Now → Request → Requested Item → Flow Designer → Create Catalog Task → Hardware Assignment

The project demonstrates how ServiceNow automation can reduce manual activities and provide a consistent workflow for standard IT equipment requests.

1.2 Purpose

The purpose of this project is to develop an automated ServiceNow workflow for standard laptop procurement.

The major objectives are:

- To provide a Standard Laptop item through the Service Catalog.
- To allow employees to submit laptop requests easily.
- To automatically trigger a Flow Designer workflow when a request is submitted.
- To automatically create a Catalog Task for the laptop request.
- To assign the task to the **Hardware** group.
- To automatically populate the required task information.
- To reduce manual intervention in the procurement process.
- To improve consistency and traceability of laptop requests.
- To simplify the overall IT procurement workflow.

2. IDEATION PHASE

2.1 Problem Statement

Organizations frequently receive requests from employees for IT equipment such as laptops. When these requests are processed manually, IT teams may have to perform several repetitive activities, including reviewing requests, creating tasks, assigning tasks, and updating request information.

The absence of an automated workflow can result in:

- Increased manual effort.
- Delays in processing requests.
- Incorrect or incomplete task information.
- Inconsistent assignment of requests.
- Difficulty tracking the progress of requests.
- Repetitive administrative work for IT support teams.

Problem Statement

How can the standard laptop procurement process be automated in ServiceNow so that a user's laptop request automatically generates the required fulfillment task and assigns it to the appropriate Hardware team?

The project addresses this problem using **ServiceNow Service Catalog and Flow Designer**.

2.2 Empathy Map Canvas

The empathy map identifies the needs and experiences of the main users involved in the laptop procurement process.

Category	User Perspective
Says	"I need a standard laptop for my work."
Thinks	"I want my request to be processed quickly and correctly."
Does	Opens the Service Catalog, selects Standard Laptop, and submits the request.
Feels	Wants a simple request process and visibility into the request status.
Pain Points	Manual request processing, delays, lack of visibility, repeated communication.
Needs	Simple request submission, automatic processing, correct task assignment, and status tracking.

IT/Hardware Team Perspective

Category	User Perspective
Says	"Laptop requests should reach the Hardware team with the required information."
Thinks	"Manual task creation takes unnecessary time."

Does	Reviews requests and performs laptop fulfillment activities.
Feels	Needs consistent and organized task assignment.
Pain Points	Repetitive task creation and incorrect/incomplete request information.
Needs	Automatically generated and correctly assigned Catalog Tasks.

2.3 Brainstorming

During brainstorming, several possible approaches were considered for improving the laptop procurement process.

Ideas Considered

1. Manual email-based laptop requests.
2. Manual ServiceNow request processing.
3. Service Catalog-based laptop request.
4. Automated task creation using Flow Designer.
5. Automated notifications for request status.
6. Automatic assignment of laptop requests to the Hardware team.
7. Automated approval handling.
8. End-to-end procurement workflow automation.

Selected Idea

The selected solution was to use **ServiceNow Service Catalog with Flow Designer** because it provides a structured request interface and allows workflow activities to be automated without requiring extensive custom programming.

3. REQUIREMENT ANALYSIS 3.1 Customer Journey Map

Stage	User Activity	System Activity	User Experience
1. Identify Need	Employee needs a laptop	—	Needs an easy way to request equipment
2. Open Catalog	Employee opens Service Catalog	Displays available catalog items	Can view available services
3. Select Item	Selects Standard Laptop	Displays laptop request	Simple selection
4. Submit Request	Clicks Order Now	Creates Request and Requested Item	Request is submitted
5. Automation	Waits for processing	Flow Designer is triggered	No manual intervention required

6. Task Creation	—	Catalog Task is automatically created	Request progresses
7. Assignment	—	Task is assigned to Hardware	Correct team receives task
8. Fulfillment	Hardware team processes task	Task status is updated	Request can be tracked
9. Completion	Employee receives laptop	Request reaches completion	Procurement process completed

3.2 Solution Requirement

Functional Requirements

FR-01: Service Catalog Item

The system shall provide a **Standard Laptop** catalog item under the Hardware category.

FR-02: Request Submission

The user shall be able to select the Standard Laptop item and submit the request using **Order Now**.

FR-03: Flow Trigger

The ServiceNow Flow Designer flow shall be triggered when the Service Catalog request is submitted.

FR-04: Catalog Task Creation

The flow shall automatically create a Catalog Task for the Requested Item.

FR-05: Short Description

The Catalog Task shall contain:

Laptop need to Configured

FR-06: Description

The Description field shall contain:

Laptop need to Configured

FR-07: Assignment Group

The Catalog Task shall be assigned to:

Hardware

FR-08: Approval

The Approval field shall be configured as:

Approved

FR-09: Request Tracking

The user shall be able to view the Request, Requested Item, and associated Catalog Task.

NonFunctional Requirements

Requirement	Description
Usability	The request process should be simple and easy to understand.
Reliability	The flow should execute consistently for Standard Laptop requests.
Maintainability	Flow configuration should be easy for ServiceNow administrators to maintain.
Automation	Manual Catalog Task creation should be minimized.
Traceability	Requests and tasks should be trackable through ServiceNow records.
Performance	The flow should execute without errors or unnecessary delays.
Security	The workflow should follow ServiceNow access and role permissions.

3.3 Data Flow Diagram



3.4 Technology Stack

Technology	Purpose
ServiceNow	Main development platform
Service Catalog	Provides the Standard Laptop request item
Flow Designer	Automates the laptop procurement workflow
Catalog Item	Represents the Standard Laptop request
Requested Item (RITM)	Stores the submitted request
Catalog Task	Represents the fulfillment task
Hardware Group	Receives the automatically generated task

ServiceNow Database	Stores requests, Requested Items, and Catalog Tasks
Web Browser	Used to access and operate the ServiceNow instance

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed solution directly addresses the identified problems in manual laptop procurement.

Problem	**Proposed Solution**
Manual request processing	Service Catalog
Manual task creation	Flow Designer
Incorrect task assignment	Automatic Hardware group assignment
Repetitive data entry	Automatic field population
Lack of tracking	ServiceNow Request/RITM/Catalog Task records
Processing delays	Automated workflow
Inconsistent task information	Predefined task values

The solution provides a structured and automated method for processing standard laptop requests.

4.2 Proposed Solution

The proposed solution consists of the following components:

Step 1: Standard Laptop Catalog Item

A **Standard Laptop** catalog item is created under the **Hardware** category.

Step 2: User Request

The employee opens the Standard Laptop catalog item and selects **Order Now**.

Step 3: Request Creation

ServiceNow creates a Request and corresponding Requested Item.

Step 4: Flow Trigger

The Service Catalog request triggers the **Standard Laptop Task** flow.

Step 5: Catalog Task Creation

The Flow Designer uses the **Create Catalog Task** action.

Step 6: Task Configuration

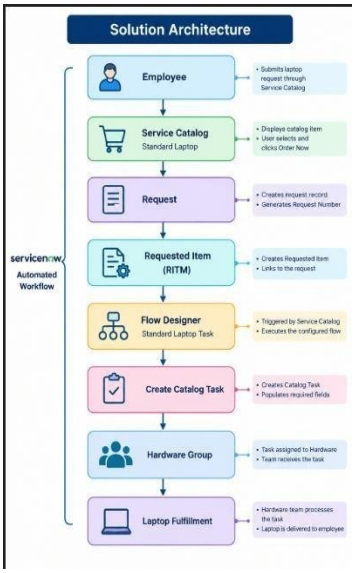
The task is populated with:

- Short Description: **Laptop need to Configured**
- Description: **Laptop need to Configured**
- Assignment Group: **Hardware**
- Approval: **Approved**

Step 7: Fulfillment

The Hardware team receives the Catalog Task and processes the laptop request. **4.3**

Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activity	Deliverable
Phase 1	Requirement Analysis	Project requirements
Phase 2	Problem Identification	Problem statement
Phase 3	Ideation	Proposed automation approach
Phase 4	ServiceNow Setup	Development environment
Phase 5	Catalog Configuration	Standard Laptop catalog item
Phase 6	Flow Development	Standard Laptop Task flow
Phase 7	Integration/Configuration	Catalog Item connected to Flow
Phase 8	Functional Testing	Test results
Phase 9	UAT	UAT report
Phase 10	Documentation	Final project report

Development Workflow



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test Case ID	Test Scenario	Expected Result	Result
FT-01	Open Standard Laptop	Catalog item should open	Pass
FT-02	Submit laptop request	Request should be created	Pass
FT-03	Verify Requested Item	RITM should be generated	Pass
FT-04	Verify Flow trigger	Flow should execute	Pass
FT-05	Create Catalog Task	Task should be created automatically	Pass
FT-06	Verify Short Description	Correct description should be displayed	Pass
FT-07	Verify Assignment Group	Hardware should be assigned	Pass
FT-08	Verify Approval	Approval should be Approved	Pass
FT-09	Verify Catalog Task	Task should appear under Requested Item	Pass
FT-10	End-to-End Test	Complete workflow should execute successfully	Pass

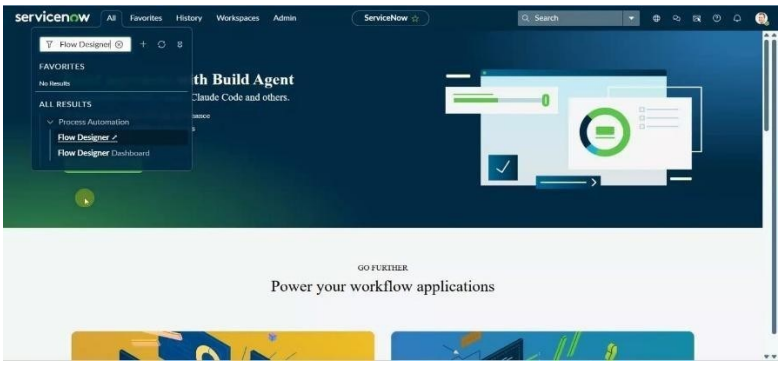
Defect Test

Severity	Number of Defects
Severity 1	0
Severity 2	0
Severity 3	0
Severity 4	0
Total Defects	**0**

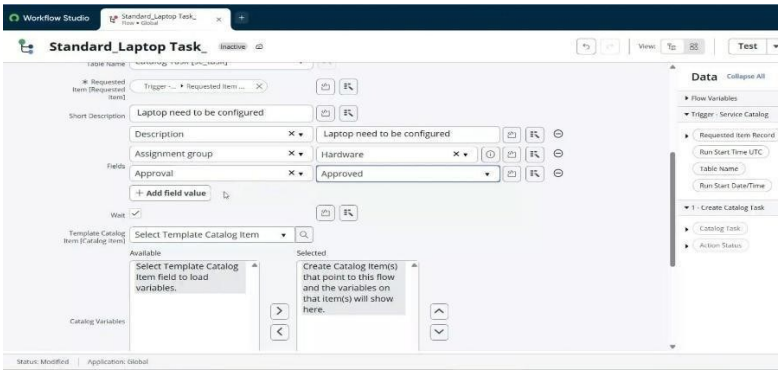
7. RESULTS

7.1 Output Screenshots

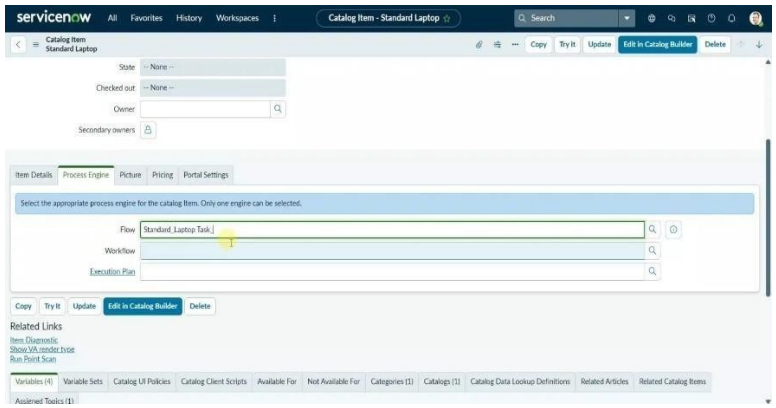
Screenshot 1 – ServiceNow All Menu



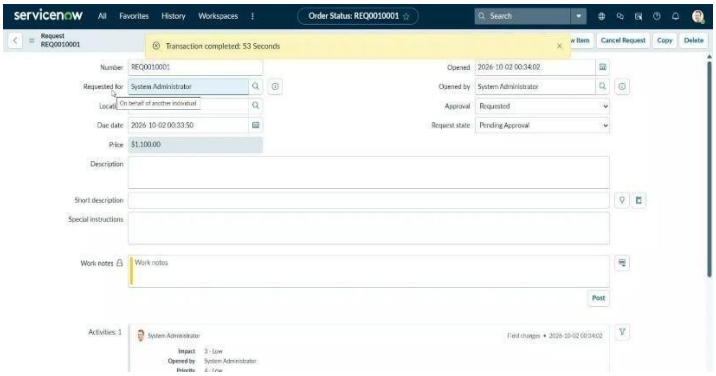
Screenshot 2 – Standard Laptop Catalog Item

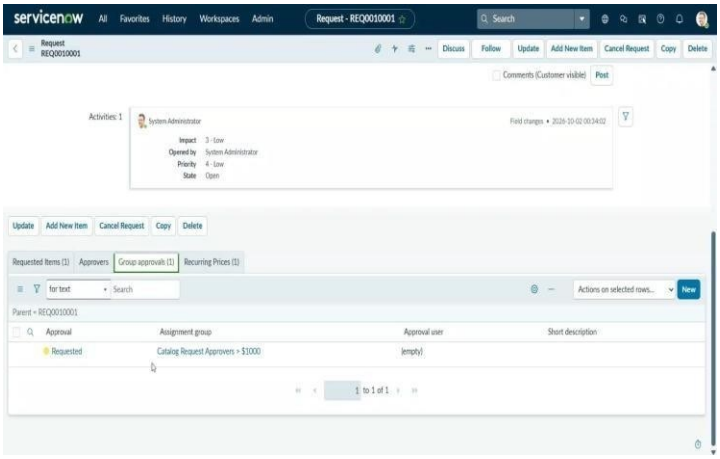


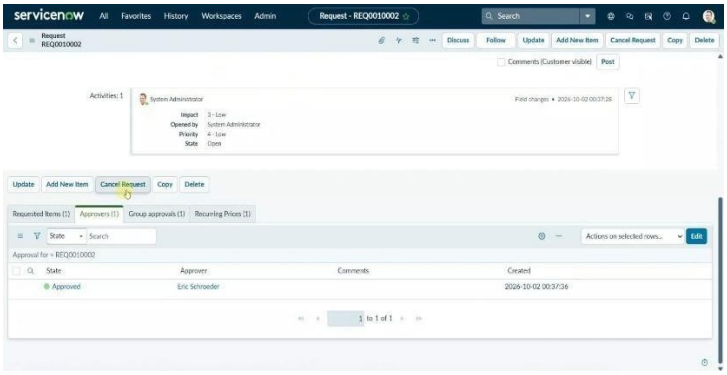
Screenshot 3 – Flow Designer



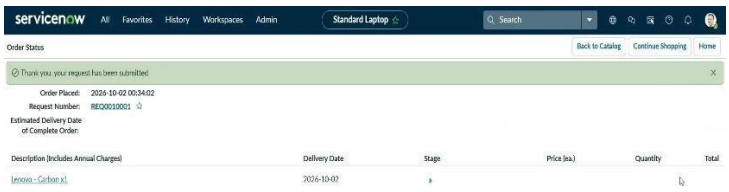
Screenshot 4 – Create Catalog Task Configuration



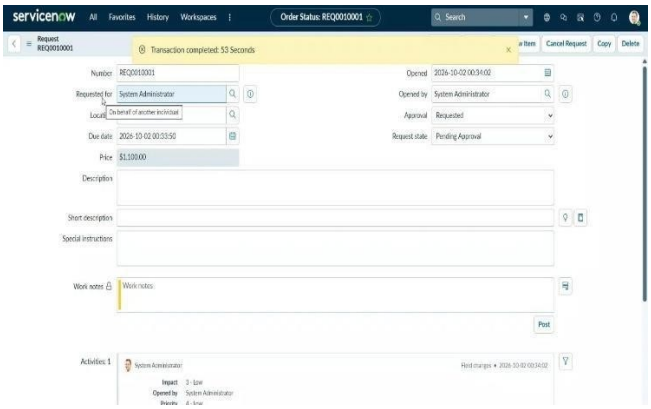




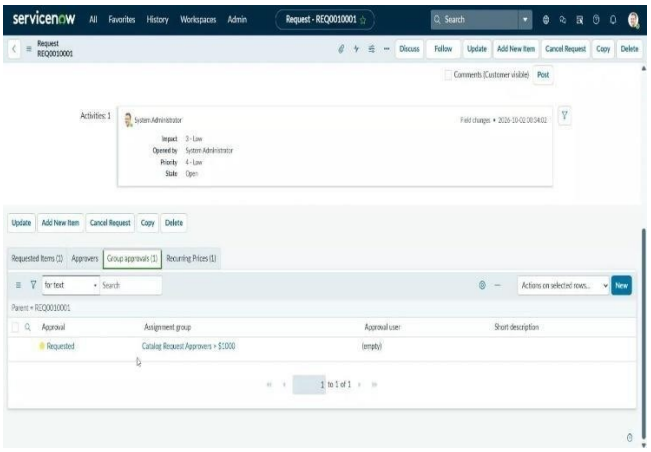
Screenshot 5 – Order Now



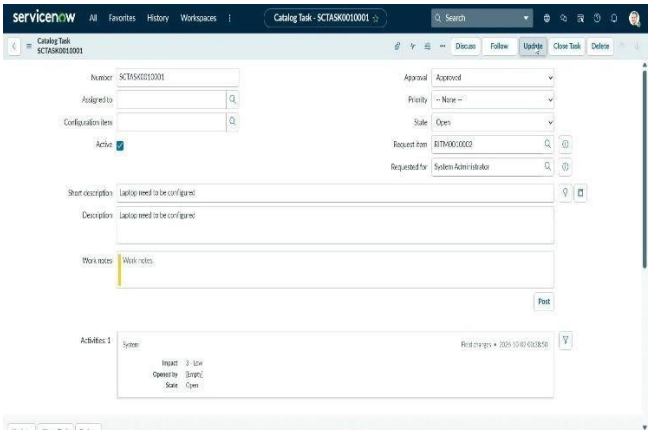
Screenshot 6 – Request Number



Screenshot 7 – Requested Item



Screenshot 8 – Catalog Task



Result Summary

The project successfully demonstrates automated Standard Laptop procurement using ServiceNow Flow Designer. The Service Catalog request triggers the configured flow, which automatically creates a Catalog Task and assigns it to the Hardware group.

8. ADVANTAGES & DISADVANTAGES

Advantages

1. **Reduces manual effort** by automatically creating Catalog Tasks.

2. **Improves consistency** by using predefined task values.
3. **Automates task assignment** to the Hardware group.
4. **Simplifies request submission** through Service Catalog.
5. **Improves traceability** through Request, Requested Item, and Catalog Task records.
6. **Reduces repetitive administrative work** for IT teams.
7. **Easy to maintain** using Flow Designer's visual workflow interface.
8. **Improves process standardization** for standard laptop requests.

Disadvantages

1. The current workflow is primarily designed for **standard laptop requests**.
2. Changes in procurement policies may require flow configuration updates.
3. The workflow depends on correct ServiceNow configuration.
4. Incorrect catalog item or flow configuration can affect task creation.
5. More complex procurement scenarios may require additional flows, approvals, or integrations.
6. External procurement systems are not included in the basic implementation.

9. CONCLUSION

The **Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer** project successfully demonstrates how ServiceNow can be used to automate a standard IT procurement process.

The solution integrates the **Service Catalog** with **Flow Designer** to automate the creation of a Catalog Task whenever a Standard Laptop request is submitted. The workflow automatically populates the task information and assigns the task to the **Hardware** group.

The implemented process provides a structured workflow:

Standard Laptop Request → Request → Requested Item → Flow Designer → Catalog Task → Hardware Group

Functional and UAT testing were performed to verify the workflow. The testing covered request submission, flow execution, Catalog Task creation, task field validation, assignment, and end-to-end processing. **No defects were identified during the reported UAT testing.**

Overall, the project demonstrates the use of ServiceNow automation to reduce manual activities and standardize the processing of IT equipment requests.

10. FUTURE SCOPE

The project can be extended with additional ServiceNow capabilities.

1. **Automated Approval**

Implement manager-based approval before the laptop request proceeds to fulfillment.

2. Inventory Integration

Integrate with an IT asset or inventory system to check laptop availability automatically.

3. Asset Creation

Automatically create or update an **Asset/Configuration Item (CI)** record when a laptop is assigned to an employee.

4. Automated Notifications

Send notifications to the requester when:

- Request is submitted.
- Request is approved.
- Catalog Task is assigned.
- Laptop is ready.
- Request is completed.

5. Multiple Laptop Types

Extend the catalog to support:

- Standard Laptop
- High-Performance Laptop
- Developer Laptop
- Business Laptop

6. Conditional Approval

Different laptop types or costs could require different approval levels.

7. Procurement Integration

Integrate ServiceNow with external procurement systems to automatically initiate purchase orders.

8. SLA Management

Configure SLAs to monitor the time taken to process and fulfill laptop requests.

9. Reporting and Dashboards

Create ServiceNow reports and dashboards to monitor:

- Number of laptop requests.
- Pending requests.
- Completed requests.
- Average fulfillment time.

- Hardware team workload.

10. Automated Asset Lifecycle

Extend the solution to support the complete laptop lifecycle, including:

Request → Approval → Procurement → Assignment → Maintenance → Replacement → Retirement

11. APPENDIX GitHub Repository:

https://github.com/Streamlining-IT-Procurement-Automation

Project Demo Link:

https://drive.google.com/file/d/1P4VDlf0AN_zP1fRGG6j7BUNzTrnsG-Rt/view?usp=drivesdk